# MTB Business Lending
# LLM Underwriting & Quick App
# Complete Implementation Guide

Version 1.0 | October 2026 | Design / proof-of-concept specification | NOT an approved credit policy

Source basis: user-provided screenshots of the account and lending entity model, field mapping, underwriting strategy, decline reasons, product definitions, and earlier Query 1 / Query 2 / Query 4 design guide. The original workbook and approved policy documents were not provided; ambiguous rules are explicitly flagged for confirmation.

## 1. Executive architecture and decision boundaries

Pipeline: Excel Query 1 / Query 2 / Query 4 -> schema and relationship validation -> canonical borrower/loan model -> feature engineering -> versioned deterministic policy engine -> explanation LLM -> decision evidence and authorized review.

• The policy engine determines the simulated routing outcome; the LLM must not change it.

• Use three separate outputs: eligibility (Quick App / full underwriting / ineligible), credit assessment (candidate / decline candidate / review), and authorization (unapproved until an authorized process signs off).

• In a POC, label results SIMULATED and do not issue a binding approval, denial, adverse-action notice, or credit offer.

• A missing key, missing credit report, stale data, unverified policy, or ambiguous relationship results in DATA_REVIEW / POLICY_REVIEW, not an invented decision.

## 2. Canonical entity relationship model

| Entity | Keys / relationships | Purpose |
|---|---|---|
| Account | account_id PK | Business or individual customer; one Account to many Loans and CreditReports. |
| LLC_BI_LOAN__C | loan_id PK; account_id FK | Loan and requested product; loan to many LCResponse; optional one-to-one LoanExtension. |
| LLC_BI_LEGAL_ENTITIES__C | entity_involvement_id PK; loan_id FK; account_id FK | Junction supporting many-to-many Account-to-Loan and borrower / guarantor roles. |
| NFUSE_CREDIT_REPORT__C | credit_report_id PK; account_id FK | One Account to many reports; select correct borrower/guarantor and report date. |
| LOAN_EXTENSION__C | loan_extension_id PK; loan_id FK UNIQUE | Behavioral underwriting program and extension attributes. |
| SCOC_LCRESPONSE__C | lcresponse_id PK; loan_id FK | SBSS response records; one loan to many responses. |
| SCOC_LCRENTITY__C | lcrentity_id PK; lcresponse_id FK; account_id FK; entity_involvement_id FK | Connects scored entities to response and loan involvement. |
| SCOC_LCRDATASOURCEANALYSIS__C | analysis_id PK; lcrentity_id FK | One LCREntity to many source analyses. |
| SCOC_LCCHARACTERISTIC__C | characteristic_id PK; analysis_id FK | Key/value SBSS credit characteristics. |
| SCOC_LCINDICATOR__C | indicator_id PK; analysis_id FK | Key/value SBSS indicators, e.g. thinFileIndicator=Y. |

| Entity | Keys / relationships | Purpose |
|---|---|---|
| MTB_REGULATION__C | regulation_id PK; loan_id FK | Communicated credit decision record; historical evidence, not a model prediction. |

Important relationship nuance: The screenshot describes a direct Account-to-Loan 1:N relationship AND a many-to-many association via the legal-entities junction. Keep both: primary account on loan, and all participating borrower/guarantor accounts in the junction. Do not assume every joined row is the primary borrower.

## 3. Excel file mapping and join contract

| Input | Required columns / linkage | Transformation |
|---|---|---|
| Query 1 - master | loan_id, primary account_id, requested amount, product, product type/line, account demographics, financials, group credit, SBSS response / score when available | One canonical row per loan_id. Deduplicate account/loan and attach most relevant score by loan and response date. |
| Query 2 - indicator | analysis_id, lcrentity_id, lcresponse_id, loan_id/account_id through relationships, indicator_key, indicator_value, borrower_type | Pivot key/value per analysis or scored entity; retain raw rows, source IDs, and conflicts. |
| Query 4 - bureau | credit_report_id, account_id, report date, credit score, freeze, public records, bankruptcies, charge-offs, liens, OFAC, past due, collection | Join by account_id and select appropriate recent report per entity; never join on row order. |

Query 2 screenshots show fields including SCOC__IndicatorKey__c, SCOC__IndicatorValue__c, SCOC__SBSS7Score__c, LLC_BI__Borrower_Type__c, and relationships through DataSourceAnalysis, LCREntity and LCResponse. Query 4 uses nFUSE__Credit_Score__c, nFUSE__Credit_Report_Freeze__c, nFUSE__OFAC_Alert_Indicator__c, and related credit report fields. API names must be checked against the source workbook before deployment.

## 4. Canonical data quality and identity validation

• Check uniqueness of primary keys; verify foreign-key referential integrity across all three exports.

• Require a primary loan account; use legal-entity involvement to identify Borrower vs Guarantor and assess each relevant report independently.

• Require effective dates for report, score and underwriting event; define approved recency limits.

• Normalize numeric money, credit scores, dates, Boolean Y/N and nulls; retain raw source value and normalized value.

• Reject conflicting values for the same key at the same source timestamp; do not silently take first row.

• Document source-to-target mapping, join cardinality, null rate, duplicate count and validation outcome per loan.

## 5. Product eligibility and routing (source-derived)

| Rule / product | Source-derived condition | Proposed routing |
|---|---|---|
| Quick App | New money <= $100M and Tier I Group Credit <= $250M; NFPs excluded | Quick App eligible only if all eligibility and data checks pass. |
| Express BALOC | Unsecured; <= $100M. Increase over $100M may still use Quick App if total new money < $100M and group credit < $250M (as written in screenshot). | Route to product-specific check; clarify existing vs incremental exposure. |

| Rule / product | Source-derived condition | Proposed routing |
|---|---|---|
| Conventional BALOC | Over $100M must be secured. | Collateral review / full underwriting as required. |
| BIL | Unsecured term debt <= $100M; acceptance letter required. | Document check; potential conversion to term loan when collateral taken. |
| SBA BIL | Permanent working capital only; SBA memo and CAM process. | SBA-specific documentation and guarantee review. |
| Term Loan | May be secured or unsecured; fixed rate and repayment stipulations. | Term-loan specific underwriting and collateral check. |
| Group Credit | Tier I: BCC, TL, BIL, BALOC actual loans. Tier II examples: merchant guidance, cash-management ACH, purchasing. | Compute and evidence both group-credit amounts; clarify when Tier II overrides Tier I. |

Unit warning: The screenshots literally show $100M and $250M, and the strategy chart shows $100M. Verify whether these are truly millions or represent another system unit before coding. The document does not silently convert them.

## 6. BALOC restrictions and exception checks

• SBA and conventional BALOC cannot be changed into one another on the same line; require a new line.

• Conventional secured/unsecured and express/standard may be changed, subject to collateral retention when reducing a secured line below $100M.

• A conventional line scoring SBA may require a larger requested line for SBA treatment; this is a policy-specific note requiring confirmation.

• Line increases are restricted within 6 months of opening per product-definition screenshot; decline-reasons screenshot says increases not allowed within 12 months. Conflict: obtain policy-owner resolution; do not hard-code either as approved.

• Investment real estate company BALOC cap shown as $25M. Startup BALOC cap shown as $25M revolving; term loans not subject to this stated startup BALOC cap.

## 7. Quick App knockout criteria (from underwriting strategy image)

| Condition | Screenshot interpretation | System action for POC |
|---|---|---|
| Average personal FICO < 650 | Decline | DECLINE_CANDIDATE; confirm borrower vs guarantor score selection. |
| SBSS < 160 | Decline | DECLINE_CANDIDATE. |
| Negative EBITDA | Knockout criterion | DECLINE_CANDIDATE if approved EBITDA definition and period available. |
| Bankruptcy in last 5 years | Based on discharge date | DECLINE_CANDIDATE if verified date within policy window. |
| Applicant under 18 | Knockout criterion | DATA_REVIEW / eligibility failure; confirm business vs individual applicant. |
| Maximum delinquency public records = 1, 2, 3 | Listed as knockout; meaning of codes unspecified | POLICY_REVIEW until code definitions are provided. |

| Condition | Screenshot interpretation | System action for POC |
|---|---|---|
| SBSS 160-189; ownership < 2 years; FICO 650-759; request > $100M | Specific combined knockout shown | DECLINE_CANDIDATE only when all conditions match; confirm boundary/units. |

## 8. Quick App decision tree (source-derived, nonbinding)

Prerequisites: identity joins valid, Quick App eligibility valid, required scores/ownership/NAICS available, and knockout checks completed.

| SBSS branch | Additional tests | Strategy-chart outcome |
|---|---|---|
| <160 | None | Decline |
| 160-189 | Ownership >=2 years | SBA approval path |
| 160-189 | Ownership <2 years, FICO >=760 | SBA approval path |
| 160-189 | Ownership <2 years, FICO 650-759, requested <= $100M | SBA approval path |
| 160-189 | Ownership <2 years, FICO 650-759, requested > $100M | Decline |
| 190-199 | None | SBA approval path |
| >=200 | Ownership <2 years | SBA approval path |
| >=200 | Ownership >=2 years; high-risk NAICS | SBA guarantee path |
| >=200 | Ownership >=2 years; other NAICS | Conventional approval path |

Chart annotation: high-risk NAICS examples include 23 (Construction), 4451 (Grocery / Convenience Retailers), 45 (Retail trade / Amazon sellers), 8111 (Automotive Repair and Maintenance), 8121 (Personal Care Services), and 8123 (Drycleaning and Laundry Services). The chart states conventional approval overrides SBA nodes for vehicle-only loans. Do not treat these paths as authorized loan approvals.

## 9. Decline-reason taxonomy and data availability

| Reason family | Evidence / logic from screenshots | Status |
|---|---|---|
| Bankruptcy / derogatory | Bankruptcy indicator and discharge date; public record patterns including judgements, liens, collections. | Implement with verified report/date. |
| Collections / judgement | Business: unsatisfied judgements or recent collections; screenshot notes manual review for some small older collections. Guarantor: unsatisfied judgements or collection indicator. | Needs exact age/amount thresholds. |
| Insufficient capital | Requested line relative to revenue; example text: 10% of revenue may drive line size. | Underwriter review; not a universal threshold. |
| Insufficient financial condition | YOY revenue decrease, expense increase, shrinking profits, negative EBITDA. | Implement when periods defined. |
| Limited credit experience | Accounts < 3 or thin-file indicator true. | Review / decline candidate by approved policy. |
| Delinquency / credit | Recent delinquencies, slow payments, credit concentrations, tax liens. | Requires bureau definitions and thresholds. |

| Reason family | Evidence / logic from screenshots | Status |
|---|---|---|
| Unable to verify identity / income | New-to-bank identity or income cannot be verified. | DATA_REVIEW / authorized underwriting. |
| Collateral / loan terms | Insufficient collateral, outside footprint, asset age vs loan term. | Manual review / out of POC where indicated. |
| Out of scope for POC | Negative news, lawsuits, garnishment, SBA eligibility, LTV and other screenshot-marked categories. | Report NOT_EVALUATED; never claim clear. |

Other screenshot-listed decline reasons include excessive debt obligations, insufficient length of operations (<2 years), slow personal credit pay, foreclosure/repo, broker-referred applications, business model concerns, inadequate bank-line paydown, and loan-purpose concerns. Several are marked out of scope or not currently available in the system.

## 10. Rule ordering, precedence, and policy provenance

• Stage 0: schema, foreign keys, source freshness and borrower/guarantor mapping.

• Stage 1: policy approval status and POC mode; production cannot proceed without signed, versioned policy.

• Stage 2: product/Quick App eligibility and document completeness.

• Stage 3: hard knockouts (OFAC flag must trigger compliance review/hold; do not equate an unverified match with confirmed sanction), credit freeze, legal eligibility and other verified stop conditions.

• Stage 4: credit/financial scoring, ownership and NAICS decision tree.

• Stage 5: route to SIMULATED_APPROVAL_CANDIDATE, SIMULATED_DECLINE_CANDIDATE, MANUAL_REVIEW, FULL_UNDERWRITING or DATA_REVIEW.

• Stage 6: generate explanation with evidence references and missing-input list.

Policy conflicts: Original draft guide proposed OFAC -> DENY, credit freeze -> MANUAL_REVIEW, multiple bankruptcies -> DENY/REVIEW, high past due -> REVIEW and strong SBSS -> APPROVE candidate. The newer screenshots provide more detailed decision criteria. Resolve conflicts with policy/compliance before production; a possible OFAC match is a compliance hold pending confirmation, not an automatically substantiated denial.

## 11. Rule catalog: implementation examples

| Rule ID | Input | Test / action |
|---|---|---|
| MTB-IDENTITY | account_id, loan_id, entity involvement, borrower role | Unresolved join or ambiguous borrower -> DATA_REVIEW. |
| MTB-ELIG-QUICK | requested_amount, tier1_group_credit, nonprofit | Quick App only if all source criteria satisfied. |
| MTB-KO-FICO | personal_fico | <650 -> simulated decline candidate. |
| MTB-KO-SBSS | sbss_score | <160 -> simulated decline candidate. |
| MTB-KO-EBITDA | ebitda | Negative -> simulated decline candidate. |
| MTB-KO-BANKRUPTCY | discharge_date, evaluation_date | Within 5 years -> simulated decline candidate. |
| MTB-STRATEGY | sbss, ownership_years, fico, request, naics | Apply Section 8 decision tree. |

| Rule ID | Input | Test / action |
|---|---|---|
| MTB-FREEZE | credit_report_freeze | Hold for review; cannot treat absent report as clean. |
| MTB-OFAC | ofac_alert | Potential match -> compliance review; verified status required. |
| MTB-POLICY-SIGNOFF | policy_version, owner approval, effective date | No operational authorization without approval. |

## 12. Expected machine-readable decision contract

Recommended response example (synthetic):

```
{
  "loan_id": "LOAN-0001",
  "account_id": "ACC-0001",
  "mode": "SIMULATION",
  "eligibility_route": "QUICK_APP",
  "policy_route": "SIMULATED_APPROVAL_CANDIDATE",
  "authorized_decision": "NOT_ISSUED",
  "policy_version": "DRAFT-POC-1",
  "rule_results": [
    {"id":"MTB-KO-FICO","status":"PASS","value":735},
    {"id":"MTB-STRATEGY","status":"MATCH","branch":"SBSS>=200"}
  ],
  "missing_inputs": [],
  "source_records": ["Q1:LOAN-0001", "Q2:IND-0001", "Q4:CR-0001"],
  "human_review_required": true,
  "explanation": "Synthetic test case; not a credit offer."
}
```

## 13. LLM responsibilities, prompt controls and auditability

• Supply only the canonical structured record and deterministic rule outcomes to the model, not raw unvalidated spreadsheets as policy authority.

• Prompt: explain triggered and non-triggered rules with factual evidence, identify missing information, and never invent thresholds, scores, relationships or approval authority.

• Constrain JSON output with a schema; reject mismatched route or rule IDs and retry once, otherwise fall back to a deterministic explanation.

• Store policy version, input hash, report timestamps, rule evaluations, LLM model/version, prompt version, reviewer ID and decision history.

• Separate adverse-action reason codes from free-text narrative; legally required notices and fair-lending controls need compliance review.

## 14. Validation / acceptance tests

| Test | Expected result |
|---|---|
| Strong SBSS, ownership >=2 years, low-risk NAICS, no blockers | SIMULATED_APPROVAL_CANDIDATE; no authorized approval. |
| FICO 640 | SIMULATED_DECLINE_CANDIDATE with knockout evidence. |
| SBSS 155 | SIMULATED_DECLINE_CANDIDATE. |
| Credit freeze Y | MANUAL_REVIEW / unable to evaluate score reliably. |

| Test | Expected result |
|---|---|
| Unresolved borrower vs guarantor join | DATA_REVIEW with exact missing foreign keys. |
| OFAC alert Y | COMPLIANCE_REVIEW, not automatic confirmed sanction. |
| Ownership 1 year, SBSS 175, FICO 700, amount over chart limit | Simulated decline branch, if units and rules confirmed. |
| Nonprofit Quick App | FULL_UNDERWRITING / not eligible for Quick App. |
| Policy draft / unsigned | Simulation allowed only; operational authorization blocked. |
| Conflicting BALOC increase windows | POLICY_REVIEW, not silently chosen threshold. |

## 15. Decisions required before production

• Confirm all field API names and provide the original MTB Background Info workbook, not screenshots alone.

• Resolve money units ($100M, $250M), requested amount definition, and Tier I vs Tier II exposure calculation.

• Resolve BALOC line-increase 6-month vs 12-month conflict.

• Confirm personal FICO aggregation (average across which applicants/guarantors), SBSS response selection and report recency.

• Define the meaning of delinquency public-record codes 1/2/3, negative EBITDA timeframe, collections thresholds and bankruptcy date logic.

• Approve NAICS list, vehicle-only override, SBA guarantee criteria, decline-reason mappings and compliance handling.

• Obtain formal credit-policy, compliance, model-risk and legal sign-off; validate performance, fairness and explanation quality before any live decision.

Document status: Detailed implementation guideline reconstructed from supplied screenshots. It is not a substitute for the approved underwriting policy, system-of-record schema, or formal sign-off.